

CIS220: Database Concepts

Developers: Andrew Brown,
Wolfgang Snowman.



Table of Contents

About Us	1
○ Introduction: Who We Are	
○ Client Information	
Objectives / Niche	2
○ Problem Statement: The Gap in the Current Market	
○ Objectives: How Foods Stuffs Fills the Niche	
User Expectations	3
○ Scope	
○ Description of Use Cases	
○ Primary Use Case: Building a Custom Meal	
Final Assessment	4-5
○ Summary of Objectives, and Expectations.	
○ Citation	
Diagrams	6
○ Business rules	
○ Entity Relationship Diagram	
Database Squema	7
Contributions	8
	9

About Us

Mission Statement:

Our mission is simple here at FoodStuffs™; we want to provide all the nutritional benefits of meal prep, minus the hassle by taking all the responsibility of finding fresh, nutritious ingredients for the consumer. We aim to provide a detailed database of nutritious ingredients, with helpfully provided macros. All the customer needs to do is provide us with a recipe made up of our ingredients. You give us a recipe, and we'll ensure you meet your goals!

Where are we located: Foodstuffs is a small family owned ghost kitchen (FICTIONAL) located and based in Dryden, NY. We pride ourselves in providing an easy-to-use solution for food prep and nutritional restrictions for all.

Primary Target demographic: Our primary demographic is any individual hoping to take their health into their own hands, but don't have the time to do it on their own, and require a greater level of control than what is provided by other meal prep services. You provide the recipe, we make it happen!

Anecdotes:

- Health Conscious: If you're particular in which macronutrients are right for you, just specify whatever interests you seek, and we will provide the best choices of foods, along with every macro, and their amounts listed alongside them!
- Dietary restricted: If you have health restrictions, all users have the ability to find foods based on any chosen blacklisted ingredient!
- Individuals with food allergies or intolerances: We allow for user-specified ingredients, including any type of substitute for every meal!
- Busy professionals seeking convenient, healthy, and customized meal options: We allow for all users to save any particular meal/food item they wish!

Objectives / Niche

Problem Statement: Let's admit it, food preparation is hard, and the vast majority of the market caters to fast, yet nutritionally sparse food. Fast-casual health chains offer some a partial solution to this problem, but they often can't, or don't, account for dietary restrictions. This leaves many with the inevitable conclusion that the only solution is meal prep, which is often a time consuming process. For some, it's much easier to pick fast food options just to avoid the hassle, for others they're left with no solution at all. This adds to a growing problem for society: How can we get nutritious food, without all the prep?

Goals:

- Transparency
- Easily available information
- Ease of use

Lack Of Transparency:

- Market: Nutritional information is too often, sadly lacking. Organizations put little effort into providing information on what their food consists of, and it's not unusual for a restaurant to fail to provide accurate descriptions of a dish's macro nutrients.
- Us: Our indepth database makes it easy for you to provide us with a recipe, and for us to make the necessary changes so that it fits your goals. With us in town, missing macro goals can be a thing of the past.

Lack of suitable information:

- Market: It's often too inconvenient for users to have to scour across publicly available information chasing a fleeting chance that a restaurant provides dishes that fit their dietary restrictions or allergies.
- Us: We do it better, we allow for concise customization of both food ingredients, real time restrictions can be listed out in the available forums and we make an extensive effort to provide what's right for you.

Easy To Use Nature:

- Us: The design of our organization makes the solutions to these problems a trivial task! Our individual focused nature allows us to easily make changes, and work with consumers directly!

User Expectations

Scope:

What You Can Do with Foods Stuffs:

- Browse a simple menu of meal bases (e.g., bowls, salads, wraps).
- Specify your health and taste preferences (e.g., "more protein," "less carbs," "no dairy").
- Get a customized menu to choose from any meal, or any specific food item.
- See every ingredient, along with its nutritional data, in your meal, and make changes to any item to fit your needs – including substitutions, removals, or insertions of any, or all ingredients.
- Watch your meal's nutrition facts and price update instantly as you make changes.
- Review a clear summary of your final order before checkout, which includes all nutritional data.

Description of Use Cases:

For the Health-Conscious User:

- You want a quick lunch that fits your specific macros (high-protein, keto-friendly, etc.) without estimating based on internet hear-say.

For the User with Dietary Restrictions:

- You need to avoid certain ingredients (allergies, intolerances) and want to easily build a safe, delicious meal without searching through thousands of online descriptions.

For the Particular Eater:

- You know exactly what you like and dislike, and you want a straightforward way to get a meal made your way, to your specific interests.

Building a Custom Meal:

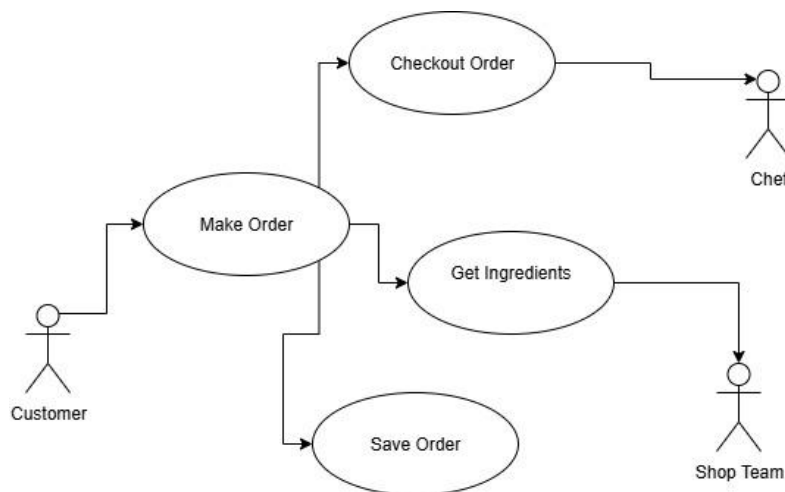
The Goal: Get a meal that matches your diet and tastes perfectly.

The Simple Process:

1. Login: You can choose to create an account for the added benefits of saving all your custom choices – from your favorite meals, to all of your health specifications, and requirements.
2. Start An Order: Choose a base for your meal to be centered around.

- (e.g., "Grain based meal", "All protein meal", ect...)
3. Specify Interests: Choose what nutritional values are most important, and which to completely remove.
(e.g., "Peanut free", "Vegan", "High in calcium", ect...)
 4. Order: Choose from a uniquely customized assortment of meals, or single food items that fit your previously specified desires.
(e.g., "Protein shake", "Loaded Sweet Potato", ect...)
 5. Customize: The system shows you all of the default ingredients for each food item, and lists their total macronutrients with all of the specifics.
 - You can remove items you don't want.
 - You can swap items for alternatives.
(e.g., brown rice for white rice, chicken for tofu)
 - You can add extra portions of items you love.
 6. See the Impact: With every change you make, the system automatically shows you the new nutrition facts and cost.
 7. Finalize: When you're happy, you can add the meal to your cart, and save the chosen meal to your very own custom meal list – which you can go back to at any time, and order the exactly specified meal you've created. You can always go back and make more changes before you check out, and you can also make changes to all your saved meals without going back through the entire order process.

Use Case Diagram:



Final Assessment

Summary:

Within the current market there is a definitive lack when it comes to easy to use, customizable, meal prep options. This creates a perfect space for a database product such as the one outlined within this document to exist. FoodStuffs perfectly meshes: meal prep, diet restrictions, and macro counting, into a simple service to provide such restrictive meals to customers. With this in mind our solution perfectly fits its niche providing ample space for improvement and growth of the project.

AI Citation

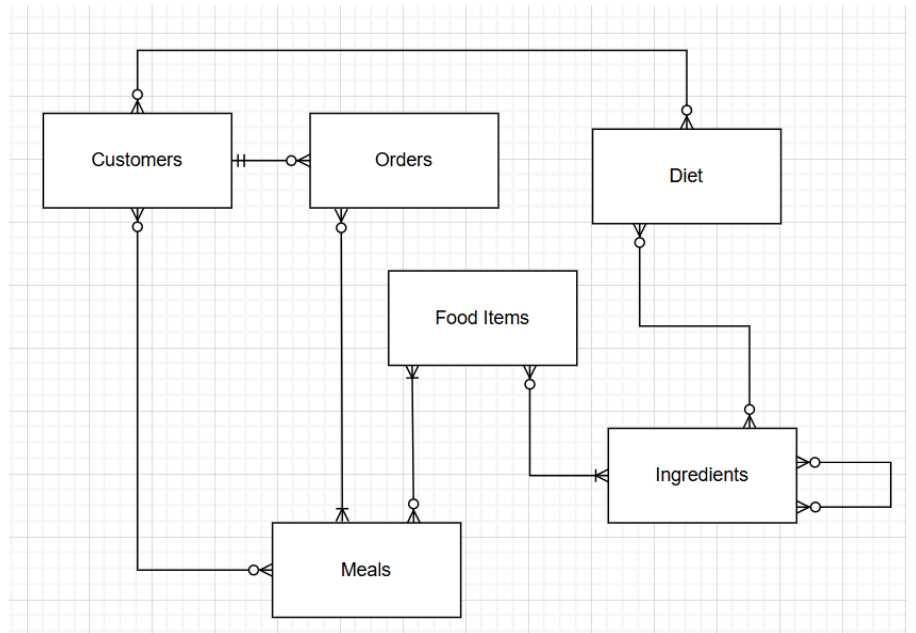
Citation:

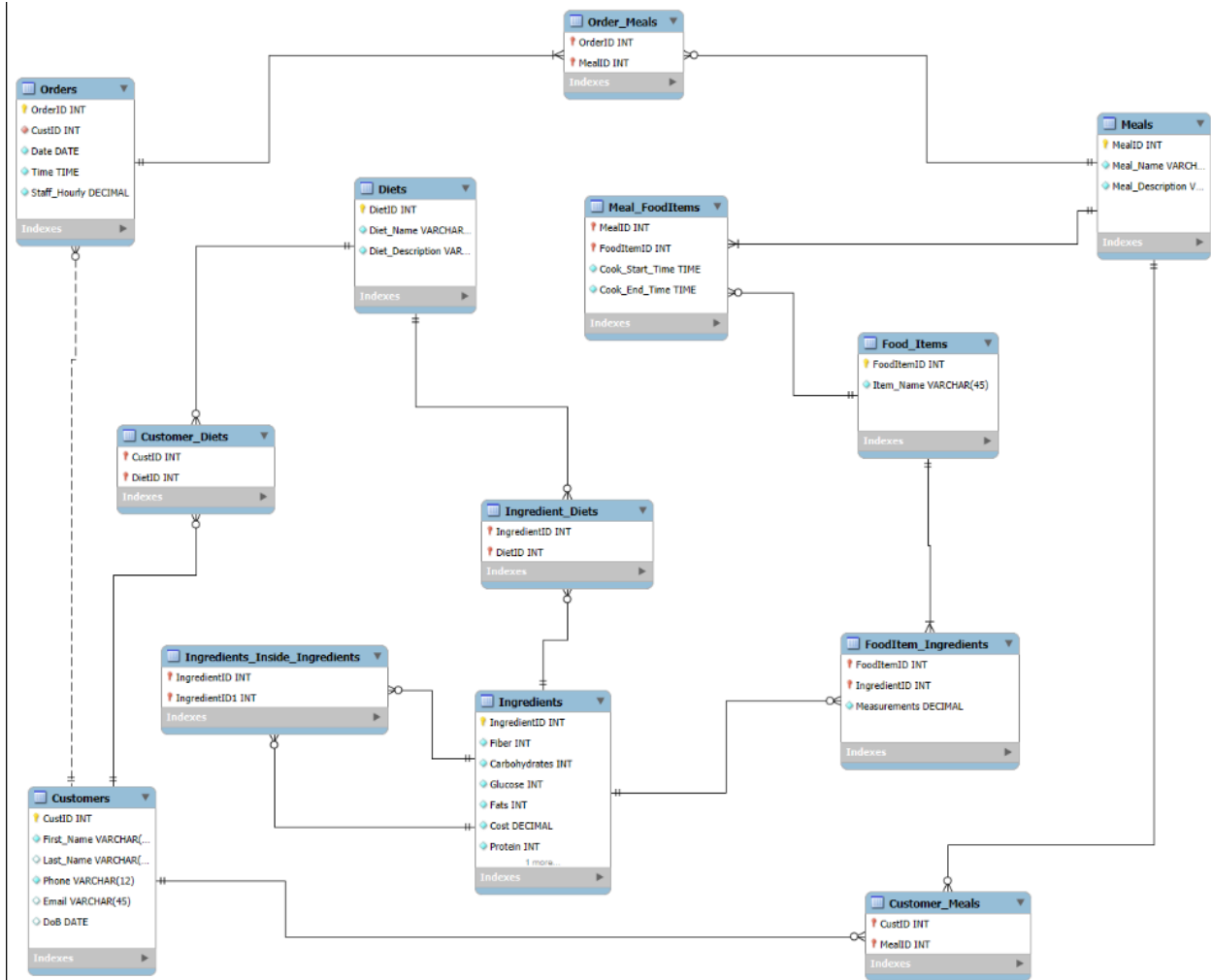
AI assistance was used for: (1) structuring initial brainstorming into a concrete document skeleton, (2) refining the table of contents, (3) adapting technical scope into user-friendly expectations, (4) expanding on original concepts, and (5) Logo. Tools included OpenAI's ChatGPT and DeepSeek's DeepSeek Chat.

A customer may have many orders.
An order must be applied to one customer.

A customer may save many meals.
A meal may be saved by many customers.

A customer may have many diets.
A diet may be applied to many customers.





Andrew Brown Contributions:

- Helped create the initial study analysis document
- Helped create every table in the database, and every attribute
- Helped design the rules of the database
- Helped transform the database into a well-designed, and working structure
- Created 3 queries
- Created the sql procedures
- Created the python script

Wolfgang Snowman Contributions:

- Helped create the initial study analysis document
- Helped create every table in the database, and every attribute
- Helped design the rules of the database
- Helped transform the database into a well-designed, and working structure
- Created 3 queries
- Helped create the sql procedures
- Helped create the python script